Caregiver Learning and Communication in Family Routines

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Caregiver-mediated communication interventions can strengthen adults' use of instructional strategies and support children's communication, but these outcomes do not necessarily improve together. Roberts et al. (2019), synthesizing 76 studies, found positive associations between parent training and caregiver and child outcomes. Their review included diverse populations and excluded studies with fewer than ten participants; it therefore supports caregiver involvement broadly rather than establishing the effects of particular mand-training procedures. More targeted studies document variable child responses despite improved caregiver implementation (McCammon et al., 2022; Suberman & Cividini-Motta, 2020).

This review asks how caregiver-mediated interventions influence functional requesting and independent communication among autistic children with emerging communication, and how caregivers' learning needs and family experiences shape implementation and the interpretation of effectiveness. Its central argument is that instructional learning, use within family life, and child communication are related but distinct. Caregiver learning needs concern the support required to understand and use procedures; family experiences help explain whether those procedures fit caregivers' priorities and daily interactions. Neither can substitute for direct evidence of child change.

This is a focused narrative review, not an exhaustive systematic review or a new analysis of participant data. Four studies anchor the comparison: Suberman and Cividini-Motta (2020), McCammon et al. (2022), Gevarter et al. (2022), and Whalon et al. (2026). Additional syntheses, caregiver-learning studies, and qualitative and methodological literature clarify their

implications. The review distinguishes study findings from the interpretive connections proposed here and from questions that require further investigation.

Research Questions and Conceptual Boundaries

Three questions for my proposed study organize the review. They concern a caregiver-training and coaching package supporting children aged three to five during family routines; the literature informs these questions without providing results for the proposed package.

RQ1. How does the caregiver-training and coaching package affect autistic children's independent requesting during family routines?

RQ2. How does the package affect caregivers' implementation accuracy during those routines?

RQ3. How do caregivers describe their experiences of using and adapting the procedures, and how do these accounts inform the interpretation of outcomes and intervention decisions?

Skinner's (1957) functional account defines a mand through its relation to motivating conditions and a characteristic consequence mediated by another person. A spoken word, gesture, manual sign, or device selection may serve as a request; its form alone does not establish its function. The proposed outcome concerns requests for access, assistance, and continuation, rather than speech production alone.

Functional requesting and independent communication are not interchangeable. Requesting is one communicative function, whereas broader communication measures may include commenting, refusing, or sharing attention. Independence concerns responding without an immediate response prompt, not communicating without AAC or a responsive partner. Likewise, an unprompted request within a familiar teaching arrangement does not establish generalization across activities or partners. These distinctions matter because the focal studies measure different outcomes, from independent mands to broader communication acts (Suberman

& Cividini-Motta, 2020; Whalon et al., 2026). The review therefore retains independent requesting as its primary child outcome without treating it as the whole of meaningful communication.

Caregiver Training and Child Requesting

Suberman and Cividini-Motta (2020) used behavioral skills training (BST) to teach caregivers mand-training procedures with speech-generating devices. Ampuero and Courtenay (2026) classify the caregiver evaluation as a nonconcurrent multiple-baseline design. Caregivers acquired the procedures, and independent child requests increased between pretraining and posttraining observations for two of three children. This provides directly relevant evidence for aided requesting, but caregiver experimental outcomes and child pre/post observations support different inferences. The latter do not independently demonstrate a functional relation for child communication.

McCammon et al. (2022) evaluated POWER using a nonconcurrent multiple-baseline design across three mother-child dyads. Mothers viewed a brief narrated video with onscreen text before conducting vocal mand training during home play. All improved implementation, although one did not reach mastery; child gains were smaller and variable. Eligibility required an echoic repertoire, limiting application to children without those skills. Follow-up was included, but transfer to new routines was not assessed. Researchers attended sessions and retained research materials between visits, so home-based performance did not establish unsupported implementation throughout daily life.

The authors' correction distinguishes Douglas and colleagues' communication-partner intervention from mand training, retracts criticism of its lack of prompting, and acknowledges POWER's structural similarity to the earlier POWR strategy. It does not alter the reported core

findings (McCammon et al., 2026). This distinction cautions against evaluating differently purposed interventions as though they shared identical requirements.

Together, the two focal studies support caregiver acquisition under specified conditions, not the superiority of one training format. Differences in participants, modalities, and procedures prevent a direct BST-versus-video comparison. More importantly, adult improvement does not resolve RQ1: children still require separate evaluation, even when caregivers learn the procedures targeted by RQ2.

Caregiver Learning Needs

Caregiver learning needs are treated here as requirements for understanding procedures, practicing them accurately, recognizing children's communication, and deciding when additional support is needed. This is an organizing interpretation, not a validated classification. It shifts attention from whether caregivers completed training to what they learned and under which support conditions they can use it.

Ampuero and Courtenay's (2026) review of 14 single-case BST studies classified ten as producing positive caregiver outcomes, including four that reached that classification after booster training. Only three examined generalization beyond the training context. The samples extended beyond preschool children. These findings make additional support part of the explanation of learning rather than an incidental detail; mastery after boosters does not demonstrate that initial instruction was sufficient.

Sone et al. (2021) found an advantage for coaching over traditional instruction on caregiver outcomes across seven early-intervention studies. In a separate analysis, parent practice with feedback was associated with increased strategy use (Sone et al., 2023). These findings support examining opportunities for practice and correction, but neither warrants assuming that

more coaching necessarily produces greater child requesting. The latter association does not isolate feedback as a causal component.

Roberts et al. (2023), in a randomized trial involving 119 mothers and autistic children, found greater postintervention use of taught responsive than directive strategies. Their findings also implicated preintervention strategy use in understanding individual differences. This suggests assessing existing practices rather than treating every caregiver as a novice. For RQ2, relevant questions include which steps require rehearsal, whether multimodal communication is recognized, and whether support remains necessary after instruction. A caregiver's difficulty using a procedure may reflect instructional needs, competing demands, or disagreement with its purpose; those possibilities require different responses rather than a single judgment of poor compliance.

Communication Support in Family Activities

Gevarter et al. (2022) studied naturalistic communication instruction in family-selected routines using a nonconcurrent multiple-probe design across three provider-parent-child triads. Children had autism or early signs of autism, and targets included gestures and manual signs. Planning, visual supports, video modeling, and coaching formed a multicomponent package. Communication turns and independent child responses increased with differing trajectories, and parent performance remained above baseline as assistance faded. The study connects instruction to family activities while extending communication beyond vocal requests. Its package design does not isolate coaching or routine selection as the cause of change.

Open-ended caregiver feedback documented implementation difficulties, demonstrating that family perspectives already contribute to this literature. Participants were English-proficient Latinx caregivers; the study does not establish equivalent results for coaching in other home

languages (Gevarter et al., 2022). Selecting a familiar routine is also not evidence of generalization. Transfer must be assessed in activities or with partners beyond the original teaching arrangement.

Whalon et al. (2026) examined Read MORe during shared reading with three autistic children aged four to six using visual scene display AAC. Their convergent mixed-methods design combined repeated observations, interviews, and weekly coaching reflections. All parents increased strategy use; two children increased communication-act frequency, whereas the third already communicated frequently in baseline. Scheduling difficulties resulted in nonconcurrent baselines, and one dyad did not complete maintenance. These conditions qualify conclusions about replicated change and durability.

Shared-reading communication acts are not equivalent to independent functional requests, and strategy frequency is not implementation accuracy. Thus, Whalon et al. (2026) offer a particularly relevant example of integration for RQ3 without supplying a direct experimental answer to RQ1 or RQ2 as defined here. The studies complement one another through different outcomes and settings rather than interchangeable demonstrations of effectiveness.

Family Experience and Intervention Use

Whalon et al. (2026) show why increased strategy use does not necessarily mean that an intervention matches family expectations. Two parents interpreted AAC displays as supporting interaction, while one viewed them as interfering and questioned the appropriateness of some strategies. Parents differed in their emphasis on following the child's lead or obtaining correct responses. Busy schedules and other daily circumstances also affected participation. These

accounts identify differences in the meaning and use of intervention; they do not prove that particular beliefs caused particular child outcomes.

Leroux et al. (2024) interviewed 37 parents of elementary-age children with complex communication needs three times during the pandemic school year. They identified similarities and differences in goals, perceptions of parental roles, and satisfaction with supports and progress. Because the sample was older and the setting pandemic-specific, this study is contextual rather than direct evidence about preschool mand training. It nevertheless shows why a shared interest in communication should not be mistaken for agreement about priorities, responsibilities, or acceptable intervention demands.

Social validity gives these judgments a distinct evidential role. Leko (2014), studying a reading intervention with teachers, illustrates how practical considerations interact in evaluations of acceptability. Applied cautiously to the caregiver literature, this perspective supports examining concrete experiences rather than relying only on favorable ratings. A parent may demonstrate accurate implementation yet find the routine burdensome, or value shared enjoyment despite limited change on a requesting measure.

The implication for RQ3 is to distinguish difficulty performing a skill from difficulty incorporating it into family life. Repeated coaching may address the former; the latter may call for reconsidering timing, targets, materials, or expectations. Caregiver accounts can guide those decisions without making families responsible for explaining every limited child gain. Family endorsement also should not replace attention to the child's engagement and refusal.

Interpreting Implementation and Child Outcomes

Caregiver performance must be interpreted in relation to the assistance available. Elmquist et al. (2025) reviewed caregiver-implemented AAC interventions and identified limited

caregiver-dependent measures and underrepresentation of marginalized families. Their review supports closer attention to how caregivers learn and implement AAC procedures, rather than treating child data as a sufficient account of the intervention. Supported practice and observations without live coaching answer different questions; success during the former cannot establish the latter.

Three levels of evidence should therefore remain distinct: whether the professional delivered the intended support, whether the caregiver used the procedures accurately, and whether the child communicated more independently. A coach-fidelity score does not measure caregiver learning. Likewise, a composite caregiver-accuracy score can conceal differences in waiting, prompting, or responding to communication. Equal overall accuracy need not mean that children encountered equivalent opportunities or consequences (McCammon et al., 2022).

Sandbank et al.'s (2023) updated Project AIM synthesis, covering 252 controlled group studies, found smaller effects for distal and generalized than proximal and context-bound outcomes, with conclusions sensitive to risk-of-bias restrictions. This is neither caregiver-specific nor a direct evaluation of mand training. Its contribution is to caution against extending a narrow observed gain into a claim about communication across contexts. Maintaining a skill, transferring it to another routine, and acquiring broader communication functions require separate evidence.

For RQ1, recording opportunities alongside independent requests would help interpret change. A higher request count might reflect more available opportunities, a greater proportion of independent responses, or both. This is a measurement implication, not a finding that opportunity frequency explains all child differences. For RQ2, reporting support and step-

specific performance would clarify what an accuracy score means. Both measures remain necessary when judging whether a family-centered package achieved its intended outcomes.

Mixed Methods and Adaptation

Qualitative description offers a way to examine experiences closely in participants' terms (Sandelowski, 2000). Analysis should preserve differing and ambivalent accounts, distinguish participants' explanations from researchers' interpretations, and consider how the coach-researcher relationship shapes disclosure. Caregiver reports of use outside recorded sessions are meaningful accounts, but they are not equivalent to direct observations of generalization.

Corr et al. (2021) identified limited reporting of recommended mixed-methods components in special education. Leko et al. (2023) provide quality indicators, while Love et al. (2023) emphasize specifying what is integrated, when integration occurs, and the relative emphasis of each approach. Together, these sources support a design in which qualitative analysis contributes more than favorable quotations appended to behavioral graphs.

Joint displays can align quantitative observations with qualitative findings and enable examination of agreement, elaboration, and divergence (Fetters et al., 2013). Whalon et al. (2026) collected reflections throughout intervention but located formal integration at the end. This distinguishes obtaining ongoing family feedback from documenting its use in decisions during intervention. For RQ3, a record linking the information available, the decision made, and subsequent observations would make that process traceable.

Adaptations also require specification. The FRAME identifies what changed, when, why, who participated, and whether modifications preserved intervention fidelity (Wiltsey Stirman et al., 2019). Changing an activity or communication form while retaining core procedures differs from altering the procedures themselves. Recording these differences prevents all departures

from being labeled either failure or appropriate tailoring. It also prevents later explanations from being mistaken for the reasons originally guiding decisions. Such records clarify implementation; they do not establish that an adaptation caused improvement. Experimental evidence remains necessary for causal claims, and concurrent changes in services or routines must remain visible in interpretation.

Synthesis and Research Positioning

The literature supports a conditional, rather than automatic, connection between caregiver intervention and child communication. Instruction can expand caregivers' skills; those skills may alter the opportunities and responses children encounter; family priorities and practical conditions influence how procedures are used. This is an interpretive synthesis of the reviewed findings, not a demonstrated mediation model. It explains why caregiver acquisition, everyday implementation, and child outcomes must be examined together without collapsing them into one indicator of success.

First, evidence for caregiver learning is often clearer than evidence for corresponding child change. BST and video-based instruction establish that caregivers can acquire procedures, but the focal mand-training studies do not show uniform improvements in independent requesting (McCammon et al., 2022; Suberman & Cividini-Motta, 2020). High implementation accuracy therefore does not justify assuming that a child's communication needs have been met. Target fit, available opportunities, modality, and continued prompting remain questions to examine, not established explanations for a particular child's response.

Second, learning needs and family experiences interact but are not synonymous. Knowing how to wait or model a response is different from finding those actions meaningful and workable in a recurring activity. Whalon et al. (2026) show that improved strategy use can

coexist with divergent judgments about appropriateness. Combined with evidence on caregiver instruction, this supports examining both learning support and the acceptability of what is being taught, rather than prescribing additional training whenever implementation is inconsistent.

Third, family accounts can change the interpretation of effectiveness. Observed gains alongside reported burden indicate benefits with constraints; favorable experiences alongside limited requesting may identify valued outcomes outside the primary measure. These are possible interpretations to test against contextualized data, not reasons to dismiss either source. The resulting synthesis should state what improved, for whom, under which conditions, and what remains uncertain.

For RQ1, the reviewed evidence justifies evaluating independent requesting across individualized modalities, while retaining clear functional and prompting criteria. My proposed study would extend the mand-training literature through repeated measures in family routines and descriptive assessment of maintenance and transfer. It would not treat increased requesting as evidence that all communication functions improved. The contribution is a bounded evaluation of an intervention package, not a claim that one approach benefits every autistic child.

For RQ2, the literature supports examining caregiver accuracy separately and interpreting it alongside the amount and type of support provided. I would build on BST and video-based studies by distinguishing uncoached probes from supported practice and documenting booster instruction and adaptations. Existing practices would inform coaching priorities rather than assuming identical learning needs. Any demonstrated functional relation would concern the package; the separate effects of video modeling, feedback, and collaborative planning would require further investigation.

For RQ3, family-selected routines, open-ended feedback, and mixed-methods shared-reading research already establish a foundation for examining family perspectives. My study would complement this work by linking caregiver accounts, behavioral observations, and dated intervention decisions during implementation as well as at final interpretation. Qualitative description and joint displays would help explain why similar measured gains can have different practical meanings, while retaining disagreement and distinguishing prospective decisions from retrospective explanations.

Overall, caregiver-mediated interventions can support functional requesting by helping familiar adults provide relevant communication opportunities and responses. Their value depends on more than adults completing procedures correctly: the procedures must also be interpreted in relation to children's observed communication and families' experiences of using them. Caregiver learning needs identify where instructional support may be necessary; family accounts clarify priorities, feasibility, and choices. Examining these dimensions together offers a stronger answer to the review's overarching question without claiming that their causal relationships have already been established.

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